

Roll No.

TEE-503

**B. TECH. (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022**

MICROPROCESSORS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Draw and explain the hardware model and programming model of 8085 microprocessor. (CO1)

OR

- (b) Explain the function of the following pins of 8085 microprocessor : (CO1)

(i) TRAP

(ii) READY

(iii) SOD

(iv) X_1 & X_2

(v) RESET OUT

2. (a) Draw and explain the architecture of 8085 microprocessor. (CO1)

OR

- (b) (i) Explain the bus organization of microprocessor.

P. T. O.

(ii) Draw and explain the function of microprocessor based computer system. (CO1)

3. (a) Define instruction cycle, machine cycle and T states. Draw the timing diagram of the following instruction : (CO2)

MOV A, M

OR

- (b) What do you mean by interrupt in microprocessor ? Classify the interrupts available in 8085 microprocessor. (CO2)
4. (a) Write the addressing modes and size of the following instructions : (CO2)

- (i) LDAX B
- (ii) CMA
- (iii) LDA 3000H
- (iv) LXI H 3050H
- (v) RAR

OR

- (b) Write an assembly language program to find the smallest number in an array. The size of array is 10 H and array is starting from memory location 3001 H. Store the result at memory location 3051H. (CO2)
5. (a) How many machine cycles will be required in the execution of the following instructions ? Write their name also : (CO2)
- (i) MVI A, 00
 - (ii) STA 3000H
 - (iii) LDAX B

(3)

(iv) LXI H 3000H

(v) SHLD 3001H

OR

- (b) What are the differences between CALL and JLJMP instructions ? How many machine cycles are required for the execution of these instructions ? Name them.

(CO2)